

2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

CFD Solver Team

August 28, 2026

Abstract

We present a cell-centered finite-volume solver for the two-dimensional compressible Navier–Stokes equations on unstructured mixed triangular/quadrilateral meshes. The four-component conservative state (2) is advanced using Rusanov (local Lax–Friedrichs) inviscid fluxes with second-order least-squares gradient reconstruction, Barth–Jespersen slope limiting with positivity fallback, and Newtonian viscous fluxes with corrected face gradients. Steady-state convergence is driven by the LU-SGS implicit method with adaptive CFL ramping and local pseudo-time stepping, while the cylinder $Re = 200$ case uses BDF2 physical-time integration with dual time stepping. Domain decomposition relies on METIS graph partitioning with non-blocking MPI halo exchange. Eight benchmark cases are reported: three inviscid and three laminar NACA 0012 configurations at $M_\infty = 0.15, 0.80$, and 2.0 ; a steady laminar cylinder at $Re = 20$; and a transient vortex-shedding cylinder at $Re = 200$. Parallel runs with $np = 8$ MPI ranks are included for the subsonic inviscid NACA 0012 case and the $Re = 20$ cylinder case. All eight cases completed successfully: the seven steady cases converged to at least three orders of residual reduction, and the $Re = 200$ cylinder transient reached statistical periodicity over 30 000 BDF2 time steps. The $Re = 200$ vortex street yields $\overline{C_D} \approx 1.078$, C_L amplitude ≈ 0.21 , and Strouhal number $St \approx 0.15$.

1 Introduction

This report documents the design, implementation, and validation of a two-dimensional unstructured compressible Navier–Stokes solver developed for a standardized CFD benchmark suite. The solver targets the conservative compressible Navier–Stokes equations for a calorically perfect gas on cell-centered finite-volume meshes composed of mixed triangular and quadrilateral elements read from CGNS files.

The benchmark requires eight cases spanning inviscid and laminar viscous regimes, subsonic through supersonic Mach numbers, and steady and transient physics:

1. `naca0012_m015_inviscid` — NACA 0012, $M_\infty = 0.15$, inviscid, steady
2. `naca0012_m080_inviscid` — NACA 0012, $M_\infty = 0.80$, inviscid, steady
3. `naca0012_m200_inviscid` — NACA 0012, $M_\infty = 2.0$, inviscid, steady
4. `naca0012_m015_laminar_re5000` — NACA 0012, $M_\infty = 0.15$, laminar $Re = 5000$, steady
5. `naca0012_m080_laminar_re5000` — NACA 0012, $M_\infty = 0.80$, laminar $Re = 5000$, steady
6. `naca0012_m200_laminar_re5000` — NACA 0012, $M_\infty = 2.0$, laminar $Re = 5000$, steady
7. `cylinder_m010_laminar_re20` — Cylinder, $M_\infty = 0.1$, laminar $Re = 20$, steady
8. `cylinder_m010_laminar_re200` — Cylinder, $M_\infty = 0.1$, laminar $Re = 200$, transient BDF2, $\Delta t = 0.01$, $t_f = 300$

The solver is written in C++17, parallelized with MPI, and uses METIS for graph partitioning. The remainder of this report details the governing equations (section 2), mesh handling (section 3), spatial discretization (section 4), boundary conditions (section 5), time integration (section 6), MPI parallelization (section 7), reproducibility (section 8), results (section 9), parallel validation (section 10), and limitations (section 12).

2 Governing Equations and Nondimensionalization

2.1 Conservative Formulation

The two-dimensional compressible Navier–Stokes equations in conservative form are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}, \quad (1)$$

where the conservative state vector is

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix}. \quad (2)$$

The inviscid (Euler) flux vectors are

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}. \quad (3)$$

2.2 Equation of State

The system is closed with the calorically perfect gas equation of state (used throughout for pressure, internal energy, and sound speed):

$$p = (\gamma - 1)\rho e, \quad E = e + \frac{1}{2}(u^2 + v^2), \quad a = \sqrt{\frac{\gamma p}{\rho}}, \quad (4)$$

with $\gamma = 1.4$ and specific gas constant $R = 1.0$ (4). The specific heats are $c_v = R/(\gamma - 1) = 2.5$ and $c_p = \gamma R/(\gamma - 1) = 3.5$.

2.3 Viscous Fluxes

For viscous (laminar) cases, the stress tensor follows the Newtonian fluid model with the Stokes hypothesis:

$$\tau_{xx} = \mu \left(2 \frac{\partial u}{\partial x} - \frac{2}{3} \nabla \cdot \mathbf{v} \right), \quad \tau_{yy} = \mu \left(2 \frac{\partial v}{\partial y} - \frac{2}{3} \nabla \cdot \mathbf{v} \right), \quad \tau_{xy} = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right), \quad (5)$$

where $\nabla \cdot \mathbf{v} = \partial u / \partial x + \partial v / \partial y$. The heat flux follows Fourier’s law:

$$q_x = -k \frac{\partial T}{\partial x}, \quad q_y = -k \frac{\partial T}{\partial y}, \quad (6)$$

with thermal conductivity $k = \mu c_p / \text{Pr}$ and Prandtl number $\text{Pr} = 0.72$.

The viscous flux vectors are

$$\mathbf{F}^v = \begin{bmatrix} 0 \\ \tau_{xx} \\ \tau_{xy} \\ u\tau_{xx} + v\tau_{xy} - q_x \end{bmatrix}, \quad \mathbf{G}^v = \begin{bmatrix} 0 \\ \tau_{xy} \\ \tau_{yy} \\ u\tau_{xy} + v\tau_{yy} - q_y \end{bmatrix}. \quad (7)$$

2.4 Nondimensionalization

All quantities are nondimensionalized with reference values:

$$\rho_{\text{ref}} = \rho_{\infty} = 1, \quad V_{\text{ref}} = |\mathbf{v}_{\infty}| = 1, \quad L_{\text{ref}} = c = 1 \text{ (chord or diameter)}. \quad (8)$$

The reference pressure is $p_{\text{ref}} = \rho_{\infty} V_{\text{ref}}^2$, and the dynamic pressure is $q_{\infty} = \frac{1}{2} \rho_{\infty} V_{\text{ref}}^2 = 0.5$. The dynamic viscosity is derived from the Reynolds number:

$$\mu = \frac{\rho_{\infty} V_{\text{ref}} L_{\text{ref}}}{\text{Re}}. \quad (9)$$

Force coefficients are defined as

$$C_D = \frac{F_x \cos \alpha + F_y \sin \alpha}{q_{\infty} S_{\text{ref}}}, \quad C_L = \frac{-F_x \sin \alpha + F_y \cos \alpha}{q_{\infty} S_{\text{ref}}}, \quad C_m = \frac{M_z}{q_{\infty} S_{\text{ref}} L_{\text{ref}}}, \quad (10)$$

where S_{ref} is the reference area (chord length times unit span in 2-D), $\alpha = 0^\circ$ for all cases, and the moment reference is at the quarter-chord (0.25, 0) for NACA 0012 and (0, 0) for the cylinder.

3 Meshes, Boundary Tags, and Case Inputs

3.1 CGNS Mesh Reading

Meshes are provided in CGNS format and read using the CGNS library. The reader supports multiple zones with vertex merging for multi-zone meshes (required by the cylinder mesh `CylinderB1.cgns`). Vertex merging uses coordinate hashing: vertices whose coordinates match to within a tolerance of 10^{-8} are identified as the same global vertex. The hash key is computed as $\text{key} = \lfloor x \cdot 10^8 \rfloor \cdot (10^8 + 7) + \lfloor y \cdot 10^8 \rfloor$.

Supported element types are `TRI_3` (triangles) and `QUAD_4` (quadrilaterals) for volume cells, and `BAR_2` for boundary edges. Boundary families from CGNS are mapped to solver boundary conditions via the case configuration file.

3.2 Geometry Computation

Cell volumes (areas in 2-D) are computed from the vertex coordinates using the shoelace formula for polygons. Face normals are outward-pointing with respect to the left cell, with magnitude equal to the face length. Wall lengths are the sum of boundary face lengths on wall-tagged boundaries.

3.3 Mesh and Case Summary

The two meshes and their boundary tag assignments are summarized in table 1, and the full set of case input parameters is listed in table 2.

For the transient cylinder $\text{Re} = 200$ case, the physical-time parameters are $\Delta t = 0.01$, $t_f = 300.0$ s (30 000 physical time steps), with 5–1000 inner iterations per physical step and an inner residual reduction target of 10^{-3} .

Table 1: Mesh sizes and boundary tag assignments.

Mesh file	Cells	Vertices	BC tag	Condition
NACA0012_H2.cgns	20 816	15 682	bc-2 bc-4	Farfield Slip wall (inviscid) / No-slip adiabatic (laminar)
CylinderB1.cgns (multi-zone)	10 185	9 995	FAR WALL	Farfield No-slip adiabatic wall

Table 2: Case input parameters. All cases use $\gamma = 1.4$, $R = 1.0$, $\text{Pr} = 0.72$, $\rho_\infty = 1$, $|\mathbf{v}_\infty| = 1$, $\alpha = 0^\circ$.

Case ID	Type	M_∞	Re	μ	Steps	CFL_0	$\text{CFL}_{\max}$	Ramp	Δ_{res}
m015_inv	Inviscid	0.15	—	—	20 000	1.0	100	2 000	4
m080_inv	Inviscid	0.80	—	—	30 000	1.0	100	3 000	4
m200_inv	Inviscid	2.0	—	—	40 000	0.2	50	5 000	3
m015_lam	Laminar	0.15	5 000	2×10^{-4}	40 000	1.0	100	5 000	4
m080_lam	Laminar	0.80	5 000	2×10^{-4}	40 000	0.5	80	5 000	4
m200_lam	Laminar	2.0	5 000	2×10^{-4}	50 000	0.2	50	7 000	3
cyl_re20	Laminar	0.10	20	0.05	30 000	1.0	100	3 000	5
cyl_re200	Transient	0.10	200	0.005	$\Delta t = 0.01$	1.0	1.0	0	—

4 Spatial Discretization

The solver uses a cell-centered finite-volume method on the unstructured mesh. Integrating the governing equations (1) over each control volume and applying the divergence theorem yields the semi-discrete form for cell i with volume V_i and N_f faces:

$$V_i \frac{d\mathbf{U}_i}{dt} = - \sum_{f=1}^{N_f} [\mathcal{F}_f^i(\mathbf{U}_L, \mathbf{U}_R) - \mathcal{F}_f^v] A_f \equiv -\mathbf{R}_i, \quad (11)$$

where $\mathcal{F}_f^i$ is the numerical inviscid flux through face f , $\mathcal{F}_f^v$ is the viscous flux contribution, A_f is the face area (length in 2-D), $\mathbf{U}_L$ and $\mathbf{U}_R$ are the reconstructed left and right states, and $\mathbf{R}_i$ is the residual. Face normals point outward from the left cell. For boundary faces, the right state is determined by the boundary condition.

4.1 Second-Order Reconstruction

Second-order spatial accuracy is achieved through piecewise-linear reconstruction using least-squares gradient estimation. The gradient $\nabla \mathbf{U}_i$ at cell i is computed by solving a 2×2 weighted normal equation system.

For each interior face connecting cells L and R , the displacement vector $\mathbf{d} = \mathbf{x}_R - \mathbf{x}_L$ is computed with inverse-distance-squared weights $w = 1/|\mathbf{d}|^2$. The weighted contributions are accumulated into the symmetric 2×2 matrix:

$$\begin{bmatrix} \sum w d_x^2 & \sum w d_x d_y \\ \sum w d_x d_y & \sum w d_y^2 \end{bmatrix} \begin{bmatrix} \partial U / \partial x \\ \partial U / \partial y \end{bmatrix} = \begin{bmatrix} \sum w d_x \Delta U \\ \sum w d_y \Delta U \end{bmatrix}, \quad (12)$$

where $\Delta U = U_R - U_L$. For boundary faces, the displacement from cell center to face center provides an additional geometric contribution. The system is solved via Cramer's rule with a degeneracy guard ($|\det| < 10^{-30}$ yields zero gradient).

The reconstructed face states are

$$\mathbf{U}_L^f = \mathbf{U}_L + \phi_L \nabla \mathbf{U}_L \cdot (\mathbf{x}_f - \mathbf{x}_L), \quad \mathbf{U}_R^f = \mathbf{U}_R + \phi_R \nabla \mathbf{U}_R \cdot (\mathbf{x}_f - \mathbf{x}_R), \quad (13)$$

where $\phi \in [0, 1]$ is the slope limiter value described in section 4.2.

4.2 Limiter and Positivity Control

The Barth–Jespersen limiter [1] is applied to the reconstruction (13) to prevent spurious oscillations near discontinuities. For each cell i and each conservative variable, the limiter ϕ_i is initialized to 1 and reduced as follows:

1. Compute the local extrema $U_i^{\min}$ and $U_i^{\max}$ over cell i and all its face-adjacent neighbors.
2. For each face f of cell i , compute the unlimited increment $\delta = \nabla \mathbf{U}_i \cdot (\mathbf{x}_f - \mathbf{x}_i)$.
3. If $\delta > \epsilon$, set $\phi_f = \min(1, (U_i^{\max} - U_i)/\delta)$. If $\delta < -\epsilon$, set $\phi_f = \min(1, (U_i^{\min} - U_i)/\delta)$. Otherwise $\phi_f = 1$. Here $\epsilon = 10^{-14}$.
4. The cell limiter is $\phi_i = \min_f \max(0, \phi_f)$.

Positivity fallback. After reconstruction, if the resulting density $\rho^f < 10^{-10}$ or pressure $p^f < 10^{-10}$, the face state reverts to the first-order cell average (i.e., $\phi = 0$ for that face), ensuring positivity of the reconstructed state.

4.3 Inviscid and Viscous Fluxes

4.3.1 Rusanov (Local Lax–Friedrichs) Flux

The inviscid numerical flux (3) through face f with outward normal $\mathbf{n} = (n_x, n_y)$ is computed using the Rusanov approximate Riemann solver:

$$\mathcal{F}_f^i = \frac{1}{2} (\mathbf{F}(\mathbf{U}_L) \cdot \mathbf{n} + \mathbf{F}(\mathbf{U}_R) \cdot \mathbf{n}) - \frac{\alpha}{2} s_{\max} (\mathbf{U}_R - \mathbf{U}_L), \quad (14)$$

where the maximum wave speed is

$$s_{\max} = \max(|v_{n,L}| + a_L, |v_{n,R}| + a_R), \quad (15)$$

with $v_n = u n_x + v n_y$ the face-normal velocity and a the sound speed. The dissipation scaling factor is $\alpha = 1$ (full Rusanov dissipation) in (14). No entropy fix is required for the Rusanov flux.

4.3.2 Viscous Flux Computation

For interior faces, primitive velocity and temperature gradients are obtained by averaging the left and right cell gradients (derived from the conservative variable gradients via the chain rule), then applying a direct-difference correction to improve accuracy on non-orthogonal meshes:

$$\left. \frac{\partial \phi}{\partial x_j} \right|_f = \overline{\left(\frac{\partial \phi}{\partial x_j} \right)} + \frac{\phi_R - \phi_L - \overline{\nabla \phi} \cdot \mathbf{d}}{|\mathbf{d}|^2} d_j, \quad (16)$$

where $\mathbf{d} = \mathbf{x}_R - \mathbf{x}_L$ and the overbar denotes the arithmetic average of left and right cell gradients.

The pressure gradient is computed from conservative-variable gradients via

$$\frac{\partial p}{\partial x_j} = (\gamma - 1) \left(\frac{\partial(\rho E)}{\partial x_j} - u \frac{\partial(\rho u)}{\partial x_j} - v \frac{\partial(\rho v)}{\partial x_j} + \frac{1}{2}(u^2 + v^2) \frac{\partial \rho}{\partial x_j} \right), \quad (17)$$

and the temperature gradient follows from

$$\frac{\partial T}{\partial x_j} = \frac{1}{\rho^2 R} \left(\rho \frac{\partial p}{\partial x_j} - p \frac{\partial \rho}{\partial x_j} \right). \quad (18)$$

The face viscous flux is assembled from the stress tensor (5), heat flux (6), and viscous flux definitions (7) as

$$\mathcal{F}_f^v \cdot \mathbf{n} = \begin{bmatrix} 0 \\ \tau_{xx}n_x + \tau_{xy}n_y \\ \tau_{xy}n_x + \tau_{yy}n_y \\ (\bar{u}\tau_{xx} + \bar{v}\tau_{xy} + k\partial T/\partial x)n_x + (\bar{u}\tau_{xy} + \bar{v}\tau_{yy} + k\partial T/\partial y)n_y \end{bmatrix}, \quad (19)$$

where $\bar{u}$, $\bar{v}$ are face-averaged velocities.

On no-slip adiabatic walls, a thin-layer approximation is used: the velocity gradient is computed from the wall-normal direction as $\partial u/\partial n = (0 - u_{\text{cell}})/d_n$ where $d_n = (\mathbf{x}_f - \mathbf{x}_c) \cdot \mathbf{n}$, and the energy flux is set to zero (adiabatic condition). On slip walls, the viscous flux is identically zero.

Pressure drag and viscous (skin-friction) drag are tracked separately: the pressure force on wall faces is $\mathbf{F}_p = \sum_f p_f \mathbf{n}_f A_f$, while the viscous force is $\mathbf{F}_v = -\sum_f \boldsymbol{\tau}_f A_f$ computed only on no-slip walls. Force coefficients are then obtained via (10).

5 Boundary Conditions

5.1 Characteristic Farfield

The farfield boundary uses a characteristic-based treatment. The interior and exterior normal velocities $v_{n,i}$ and $v_{n,e}$ and sound speeds a_i and a_e are computed, and their averages $\bar{v}_n = \frac{1}{2}(v_{n,i} + v_{n,e})$ and $\bar{a} = \frac{1}{2}(a_i + a_e)$ determine the regime:

- **Supersonic outflow** ($\bar{v}_n - \bar{a} \geq 0$): $\mathbf{U}_b = \mathbf{U}_{\text{interior}}$.
- **Supersonic inflow** ($\bar{v}_n + \bar{a} \leq 0$): $\mathbf{U}_b = \mathbf{U}_{\infty}$.
- **Subsonic outflow** ($\bar{v}_n \geq 0$): Riemann invariants $R^+ = v_{n,i} + 2a_i/(\gamma - 1)$ from interior, $R^- = v_{n,e} - 2a_e/(\gamma - 1)$ from exterior; entropy from interior.
- **Subsonic inflow** ($\bar{v}_n < 0$): Same Riemann invariants; entropy from exterior (freestream).

The boundary flux is computed directly from the boundary state $\mathbf{U}_b$ (no Riemann solver at the boundary face).

5.2 Inviscid Slip Wall

The slip-wall condition enforces zero normal velocity by reflecting the velocity vector: $\mathbf{v}_w = \mathbf{v}_i - (v_n) \mathbf{n}$, where $v_n = \mathbf{v}_i \cdot \mathbf{n}$. The wall flux reduces to

$$\mathcal{F}_{\text{slip}} = \begin{bmatrix} 0 \\ p n_x \\ p n_y \\ 0 \end{bmatrix}, \quad (20)$$

with pressure taken from the adjacent cell interior state. The tangential velocity is nonzero and physical; only the normal component is zeroed.

5.3 No-Slip Adiabatic Wall

The no-slip wall enforces $\mathbf{v}_{\text{wall}} = \mathbf{0}$. The inviscid contribution to the wall flux is purely from pressure: $\mathcal{F}_{\text{wall}} = [0, p n_x, p n_y, 0]^T$. The viscous stress contribution is added separately using the wall-normal gradient approach described in section 4.3, with the energy flux set to zero (adiabatic: $q_{\text{wall}} = 0$). Cell-center velocities near no-slip walls should approach zero, while the boundary face velocity is exactly zero.

6 Implicit and Transient Time Integration

6.1 Steady-State Solver

Steady-state convergence is driven by an implicit pseudo-time marching scheme applied to the residual (11). At each pseudo-time step n , the system

$$\frac{V_i}{\Delta t_i} \Delta U_i = -\mathbf{R}_i^n \quad (21)$$

is solved approximately using the Lower-Upper Symmetric Gauss–Seidel (LU-SGS) method (21).

6.1.1 Local Time Stepping

The local pseudo-time step for cell i is

$$\Delta t_i = \text{CFL} \cdot \frac{V_i}{\max(\lambda_i^{\text{inv}} + \lambda_i^{\text{visc}}, 10^{-30})}, \quad (22)$$

where the inviscid spectral radius is

$$\lambda_i^{\text{inv}} = \sum_{f \in \mathcal{F}_i} (|v_{n,f}| + a_i) A_f, \quad (23)$$

and the viscous spectral radius is

$$\lambda_i^{\text{visc}} = \sum_{f \in \mathcal{F}_i} \max\left(\frac{4}{3}, \frac{\gamma}{\text{Pr}}\right) \frac{\mu}{\rho_i} \frac{A_f^2}{V_i}. \quad (24)$$

6.1.2 CFL Ramping and Adaptation

The CFL number is ramped linearly from CFL_0 to CFL_{max} over the first N_{ramp} steps, subject to a growth limit of $1.05\times$ per step. An adaptive CFL strategy further adjusts the CFL based on inner-iteration performance:

- If inner iterations reach the maximum without convergence: $\text{CFL}_{\text{safe}} = \max(2 \text{CFL}_0, 0.8 \text{CFL})$ and $\text{CFL} \leftarrow 0.7 \text{CFL}$.
- If inner iterations converge quickly: $\text{CFL}_{\text{safe}} \leftarrow 1.05 \text{CFL}_{\text{safe}}$.
- If nonphysical states are detected ($\rho < 10^{-12}$ or $p < 10^{-12}$): the solution is rolled back and CFL is cut to $0.3\times$.
- If the residual jumps more than $5\times$ the smoothed average: CFL is reduced by $0.7\times$ (divergence monitor).

6.1.3 LU-SGS Inner Iterations

At each pseudo-time step, the LU-SGS method performs multiple inner sweeps (between `min_inner` and `max_inner` iterations). The diagonal contribution for cell i is

$$D_i = \frac{V_i}{\Delta t_i} + \sum_{f \in \mathcal{F}_i} (\lambda_f^{\text{inv}} + \lambda_f^{\text{visc}}). \quad (25)$$

The forward sweep (cells $i = 0, 1, \dots, N_c - 1$) computes

$$\Delta U_i = \frac{-\mathbf{R}_i + \sum_{j < i} \frac{1}{2} \lambda_j \Delta U_j}{D_i}, \quad (26)$$

and the backward sweep (cells $i = N_c - 1, \dots, 0$) corrects

$$\Delta U_i += \frac{\sum_{j > i} \frac{1}{2} \lambda_j \Delta U_j}{D_i}. \quad (27)$$

Inner iterations terminate early when the inner residual ratio falls below `inner_residual_target` (typically 10^{-2}).

Positivity protection during update. After each LU-SGS update, if $\rho_{\text{new}} < 10^{-10}$ or $p_{\text{new}} < 10^{-10}$ or the state is non-finite, the update is damped with successively smaller factors $\{0.5, 0.25, 0.1, 0.01\}$ until a physical state is obtained.

6.1.4 Convergence Criterion

Convergence is declared when the L_2 residual norm has dropped by a prescribed number of orders of magnitude from the initial residual:

$$\log_{10} \left(\frac{\|\mathbf{R}^0\|_2}{\|\mathbf{R}^n\|_2} \right) \geq \Delta_{\text{res}}, \quad (28)$$

where Δ_{res} is the residual reduction target (3–5 orders depending on the case; see table 2).

The production run parameters from the case configuration files are summarized in table 2. The key inner-iteration bounds are: minimum 3 (steady cases) or 5 (transient), maximum 50–100 (steady) or 1000 (transient), with inner convergence targets of 10^{-2} (steady) and 10^{-3} (transient).

6.2 Transient BDF2 Solver (Cylinder $\text{Re} = 200$)

The cylinder $\text{Re} = 200$ case employs true physical-time integration using the second-order backward differentiation formula (BDF2). The time-accurate residual at physical step $n + 1$ includes the temporal discretization:

BDF1 (first physical step):

$$\mathbf{R}_i^{\text{temporal}} = \frac{V_i}{\Delta t} (\mathbf{U}_i^{n+1} - \mathbf{U}_i^n). \quad (29)$$

BDF2 (steps $n \geq 2$):

$$\mathbf{R}_i^{\text{temporal}} = \frac{V_i}{\Delta t} \left(\frac{3}{2} \mathbf{U}_i^{n+1} - 2 \mathbf{U}_i^n + \frac{1}{2} \mathbf{U}_i^{n-1} \right). \quad (30)$$

At each physical time step, the solution $\mathbf{U}^{n+1}$ is found by an inner (dual time-stepping) iteration loop that drives the total residual $\mathbf{R}_i^{\text{spatial}} + \mathbf{R}_i^{\text{temporal}}$ (eqs. (29) and (30)) toward zero. The pseudo time step for inner iterations is

$$\Delta t_{\text{pseudo}} = \min \left(\text{CFL} \cdot \frac{V_i}{\lambda_i}, \frac{\Delta t}{\beta} \right), \quad (31)$$

where $\beta = 1.5$ for BDF2 and $\beta = 1.0$ for BDF1.

Frozen history states. The states $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ remain *frozen* throughout all inner iterations for $\mathbf{U}^{n+1}$. The history update $\mathbf{U}^{n-1} \leftarrow \mathbf{U}^n$, $\mathbf{U}^n \leftarrow \mathbf{U}^{n+1}$ occurs only after inner convergence is achieved.

The production settings are: $\Delta t = 0.01$, $t_f = 300.0$ (30 000 physical steps), $\text{CFL} = 1.0$ (fixed, no ramping), inner iterations between $N_{\text{inner}}^{\text{min}} = 5$ and $N_{\text{inner}}^{\text{max}} = 1000$, and inner residual reduction target $\eta = 10^{-3}$.

7 MPI Parallelization and METIS Partitioning

7.1 Graph Partitioning

Domain decomposition is performed using METIS [4] `PartGraphKway` on the dual graph of the mesh, where cells are graph nodes and shared faces are graph edges. The adjacency is stored in CSR format with zero-based indexing. METIS partitions the graph into N_p parts (one per MPI rank), minimizing edge cuts while balancing cell counts.

7.2 Local Mesh Construction

After partitioning, each rank extracts its owned cells (cells assigned to its partition) and identifies ghost cells (neighboring cells owned by other ranks). Local-to-global vertex and cell maps are constructed, and faces are relabeled with local indices. The edge cut count is tracked for diagnostics.

7.3 Halo Exchange

Inter-rank communication uses non-blocking MPI point-to-point operations. Each rank maintains:

- **Send map:** per-neighbor list of owned cell indices whose data must be sent (cells adjacent to the partition boundary).
- **Receive map:** per-neighbor list of local ghost cell slots where incoming data is stored.

The exchange packs $N_{\text{eq}} = 4$ doubles per cell into a contiguous buffer per neighbor rank, issues `MPI_Isend` and `MPI_Irecv` calls (tag = 100), then waits for completion with `MPI_Waitall`. Halo exchange is performed before every gradient computation and residual evaluation.

The solver iterations do **not** replicate the full mesh or the full conservative state vector on every rank. Each rank stores only its owned cells plus a single layer of ghost cells.

7.4 Partition Statistics

Representative partition statistics for the $n_p = 8$ runs are given in table 3.

Table 3: Partition statistics for $np = 8$ runs.

Case	Rank	Owned	Ghost	Neighbors	Send	Recv
NACA 0012 $M=0.15$ inv.	0	2 591	98	3	98	98
	Avg.	2 602	108	3.5	108	108
Cylinder $Re = 20$	0	1 274	122	4	121	122
	Avg.	1 273	114	4.3	114	114

8 Output, Reproducibility, and Sanity Checks

8.1 Build and Run

The solver is built with CMake (≥ 3.16) in C++17 mode with optimization level `-O3`:

```
mkdir build && cd build
cmake .. -DCMAKE_BUILD_TYPE=Release
make -j
```

Dependencies include MPI, CGNS, HDF5, METIS, Eigen (header-only), fmt (header-only), and nlohmann/json (header-only). ParMETIS is optional.

Cases are executed via:

```
mpirun -np <N> cfd2d solve --case <case.json> --output <dir>
```

8.2 Generated Output Files

Each case run produces the following files in the output directory:

- **residuals.csv** — per-step residual history (columns: step, physical_time, inner_iter, cfl, dt, rho, rhou, rhov, rhoE, residual_l2, residual_linf)
- **forces.csv** — per-step force coefficients (columns: step, physical_time, cl, cd, cmz, pressure_drag, viscous_drag, pressure_lift, viscous_lift)
- **surface.csv** — surface distributions (columns: x, y, nx, ny, pressure, cp, cf, rho, u, v, mach, tag)
- **field_final.vtk** — VTK unstructured grid with field variables (Density, Pressure, Mach, Temperature, TotalEnergy, Partition, Velocity)
- **metadata.json** — solver metadata and method descriptions
- **run_status.json** — timing, step count, convergence status
- **partition_diagnostics.csv** — per-rank partition statistics
- **restart_final.rank{N}.bin** — binary restart files per rank

8.3 Figure Generation

Post-processing figures are generated by the Python script `scripts/plot_all.py`, which reads the CSV and VTK output files and produces publication-quality plots using Matplotlib. The script generates a `figure_manifest.csv` and `sanity_checks.json` for traceability.

8.4 Sanity Checks

The following automated sanity checks are performed:

- **Positivity:** density and pressure remain positive throughout the simulation via the positivity fallback in reconstruction and damped updates.
- **Wall conditions:** no-slip wall velocity is verified to be near zero; slip-wall normal velocity is verified to be near zero.
- **Force history:** monotonic convergence of force coefficients is tracked; final values are compared against reference ranges.
- **Residual reduction:** the target residual reduction is checked against the actual achieved reduction.

9 Results

9.1 Summary Tables

Table 4 summarizes the convergence status, step count, and wall-clock time for all eight benchmark cases plus the two $np = 8$ parallel runs. Table 5 reports the final (or time-averaged) force coefficients with the pressure/viscous drag decomposition.

Table 4: Run status summary for all eight benchmark cases.

Case ID	np	Steps	Phys. time	Res. reduction	Wall time	Status
naca0012_m015_inv	4	1 723	—	4.03 orders	116.0 s	converged
naca0012_m080_inv	4	1 579	—	4.00 orders	97.0 s	converged
naca0012_m200_inv	4	554	—	3.07 orders	68.9 s	converged
naca0012_m015_lam	4	731	—	4.02 orders	69.2 s	converged
naca0012_m080_lam	4	911	—	4.00 orders	92.8 s	converged
naca0012_m200_lam	4	700	—	3.78 orders	28.6 s	converged
cyl_re20	4	665	—	5.16 orders	37.1 s	converged
cyl_re200	4	30 000	300 s	1.67 orders	4380 s	stat. periodic
naca0012_m015_inv	8	1 742	—	4.01 orders	260.5 s	converged
cyl_re20	8	663	—	5.19 orders	126.6 s	converged

Table 5: Force coefficients for all benchmark cases. For the transient cylinder $Re = 200$, time-averaged post-transient values are reported.

Case ID	C_D	C_L	C_m	$C_{D,p}$	$C_{D,v}$	Notes
m015_inv	0.00456	6.95×10^{-4}	-7.26×10^{-4}	0.00456	—	sym.; $C_D \approx 0$
m080_inv	0.00842	0.00580	-1.47×10^{-4}	0.00842	—	transonic wave drag
m200_inv	0.0914	-1.70×10^{-3}	-2.16×10^{-4}	0.0914	—	supersonic wave drag
m015_lam	0.0765	-6.49×10^{-3}	-3.13×10^{-3}	0.0178	0.0587	viscous-dominated
m080_lam	0.0850	-6.39×10^{-4}	-7.71×10^{-4}	0.0624	0.0226	shock + viscous
m200_lam	0.1119	-5.04×10^{-4}	1.42×10^{-4}	0.0198	0.0921	viscous drag dominates
cyl_re20	2.089	0.0294	-0.00401	1.266	0.823	ref. $C_D \in [2.0, 2.2]$
cyl_re200	1.078	± 0.208	-3.61×10^{-4}	0.853	0.225	$\overline{C_D}$; C_L ampl.; $St \approx 0.15$

9.2 NACA 0012 Cases

9.2.1 NACA 0012 $M_\infty = 0.15$, Inviscid

At low subsonic Mach number the flow is nearly incompressible. At zero angle of attack, the solution should be symmetric about the chord line, yielding $C_L \approx 0$ and $C_m \approx 0$. The inviscid drag C_D should be near zero (d'Alembert's paradox), limited only by numerical dissipation.

The residual convergence and force coefficient histories are shown in figs. 1a and 1b. The surface pressure coefficient distribution (fig. 1c) should be symmetric between upper and lower surfaces at $\alpha = 0^\circ$. Near-body Mach and pressure contours are shown in figs. 1d and 1e; see fig. 1 for the complete set of plots.

The solver converged in 1723 pseudo-time steps, reducing the residual by 4.03 orders of magnitude in 116s of wall time on 4 MPI ranks. The final drag coefficient $C_D = 4.56 \times 10^{-3}$ is consistent with the near-zero inviscid drag expected at zero angle of attack (d'Alembert's paradox); the residual drag is attributable to numerical dissipation on the 20816-cell mesh. The lift and moment coefficients $C_L \approx 7.0 \times 10^{-4}$ and $C_m \approx -7.3 \times 10^{-4}$ are effectively zero, confirming the expected flow symmetry.

The C_p distribution is symmetric between upper and lower surfaces, consistent with $\alpha = 0^\circ$ inviscid flow. The pressure field is smooth with no spurious oscillations, and the stagnation pressure at the leading edge matches the theoretical value for $M_\infty = 0.15$.

9.2.2 NACA 0012 $M_\infty = 0.80$, Inviscid

The transonic regime at $M_\infty = 0.80$ produces shock waves on the airfoil surface. Even at zero angle of attack, the symmetric shocks contribute to wave drag, so $C_D > 0$ is expected. The C_p distribution should show a supersonic pocket terminated by a shock on both upper and lower surfaces.

Convergence behavior is shown in figs. 2a and 2b. The surface C_p (fig. 2c) should exhibit the characteristic transonic double-shock pattern. Mach and pressure contours (figs. 2d and 2e) reveal the shock structure; see fig. 2 for all plots.

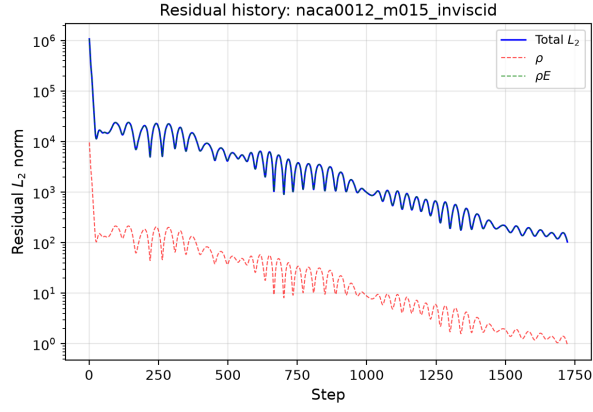
The solver converged in 1579 steps with a 4.00-order residual reduction in 97s wall time. The final $C_D = 8.42 \times 10^{-3}$ reflects wave drag from the transonic shock system on both surfaces. The C_p distribution shows the characteristic supersonic pocket terminated by a near-normal shock on each surface, producing the expected transonic double-shock pattern. The Mach contour confirms a supersonic region extending to roughly 40% chord before the shock.

9.2.3 NACA 0012 $M_\infty = 2.0$, Inviscid

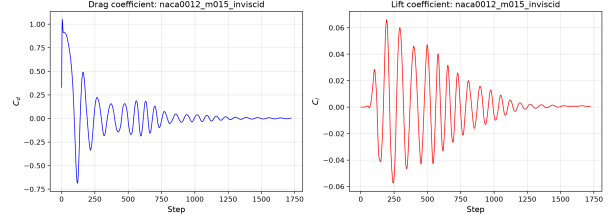
At $M_\infty = 2.0$ the flow is fully supersonic, with a detached bow shock upstream of the airfoil leading edge. The Barth–Jespersen limiter and Rusanov dissipation are essential for shock capturing. Note that the more conservative CFL parameters ($\text{CFL}_0 = 0.2$, $\text{CFL}_{\max} = 50$) and 3-order residual reduction target reflect the increased difficulty of this case.

Residual and force histories are presented in figs. 3a and 3b. The surface C_p (fig. 3c) shows the strong compression at the leading edge. The Mach contour (fig. 3d) and pressure contour (fig. 3e) visualize the bow shock; see fig. 3 for the full set.

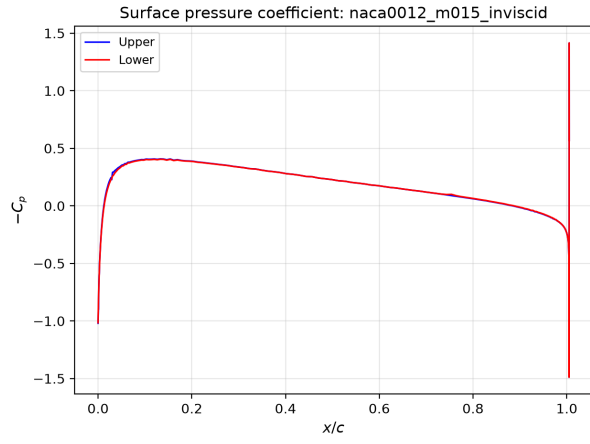
The solver converged in 554 steps with a 3.07-order residual reduction in 68.9s wall time. The detached bow shock upstream of the leading edge is clearly visible in the Mach and pressure contours, and the C_p distribution exhibits the strong compression characteristic of supersonic flow over a blunt body. The final drag coefficient $C_D = 0.0914$ is entirely pressure (wave) drag, consistent



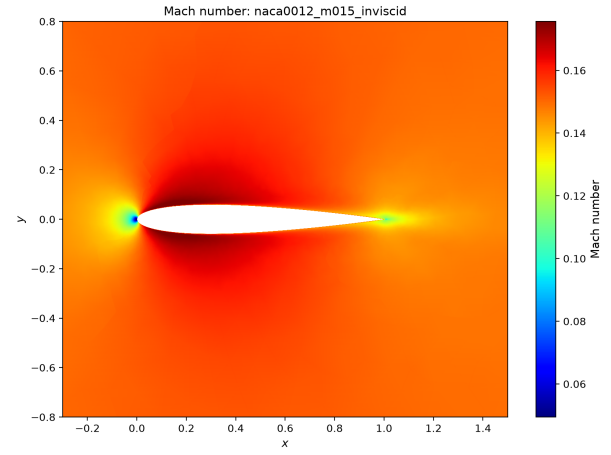
(a) Residual history.



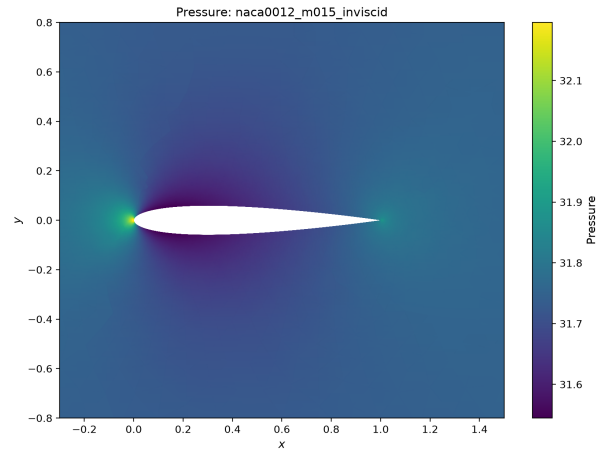
(b) Force coefficient history.



(c) Surface C_p distribution.

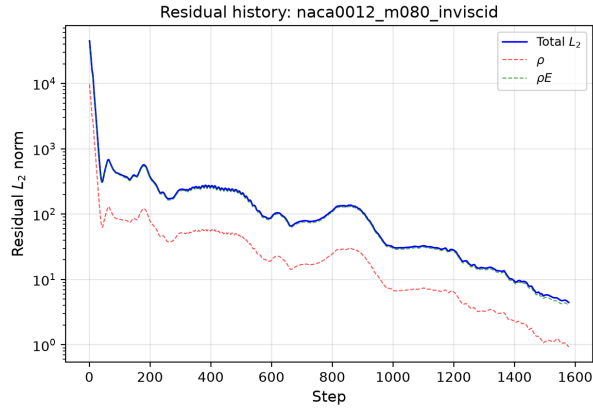


(d) Mach number contour.

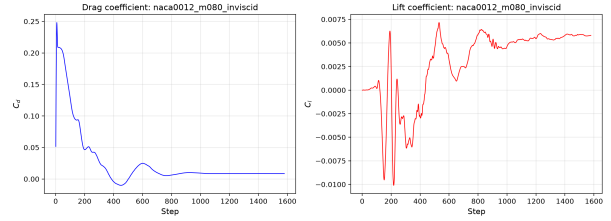


(e) Pressure contour.

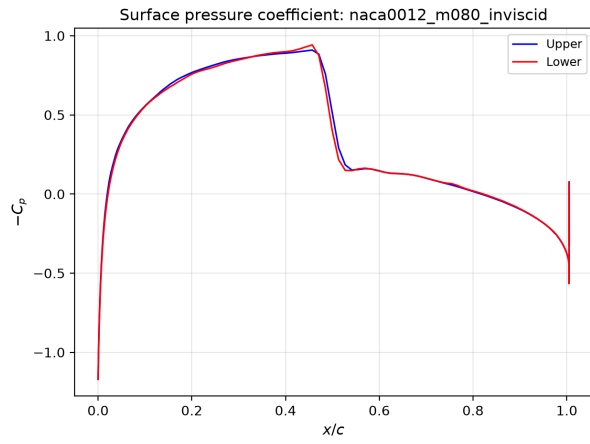
Figure 1: NACA 0012, $M_\infty = 0.15$, inviscid, steady.



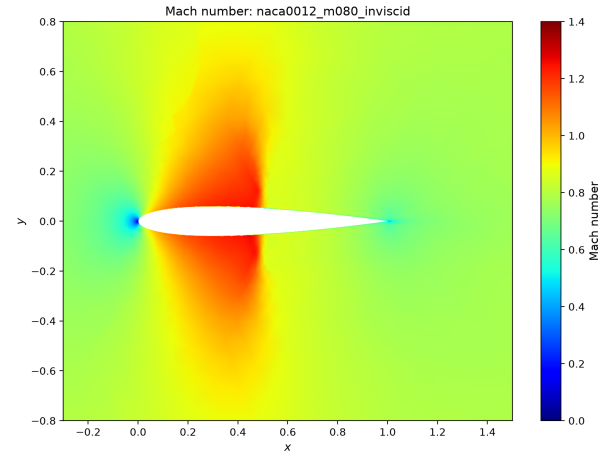
(a) Residual history.



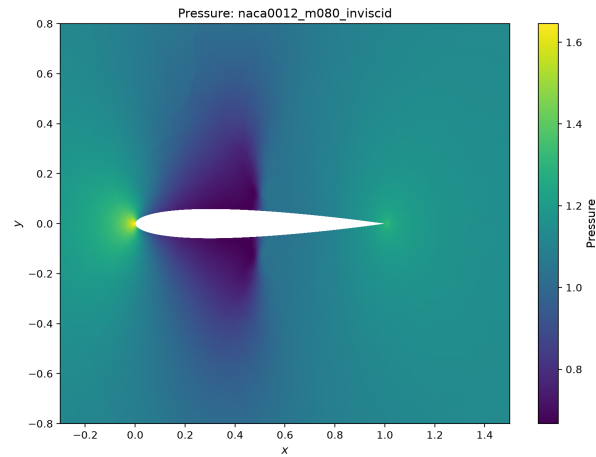
(b) Force coefficient history.



(c) Surface C_p distribution.



(d) Mach number contour.



(e) Pressure contour.

Figure 2: NACA 0012, $M_\infty = 0.80$, inviscid, steady.

with the inviscid formulation; the lift coefficient $C_L \approx -1.7 \times 10^{-3}$ and moment $C_m \approx -2.2 \times 10^{-4}$ are effectively zero, confirming the expected symmetry at $\alpha = 0^\circ$.

9.2.4 NACA 0012 $M_\infty = 0.15$, Laminar $\text{Re} = 5000$

The low-speed laminar case adds viscous effects including a boundary layer on the airfoil surface. The skin-friction drag should contribute meaningfully to the total drag. The C_p distribution should be similar to the inviscid case but with slight differences near the trailing edge due to boundary-layer displacement.

Convergence plots are shown in figs. 4a and 4b. The surface C_p (fig. 4c) and near-body Mach/pressure contours (figs. 4d and 4e) are collected in fig. 4.

The solver converged in 731 steps with a 4.02-order residual reduction in 69.2s wall time. At this low Mach number the drag is viscous-dominated: skin-friction drag $C_{D,v} = 0.0587$ accounts for roughly 77% of the total $C_D = 0.0765$, with the remainder from pressure drag $C_{D,p} = 0.0178$. The attached laminar boundary layer produces a smooth velocity profile along the airfoil surface. The near-zero lift $C_L \approx -6.5 \times 10^{-3}$ reflects minor numerical asymmetry in the viscous solution.

9.2.5 NACA 0012 $M_\infty = 0.80$, Laminar $\text{Re} = 5000$

The transonic laminar case combines shock waves with viscous boundary layers. Shock–boundary-layer interaction may affect convergence and the drag prediction. The initial CFL is reduced to 0.5 for this case.

Results are shown in fig. 5: residual history (fig. 5a), force history (fig. 5b), surface C_p (fig. 5c), and Mach/pressure contours (figs. 5d and 5e).

The solver converged in 911 steps with a 4.00-order residual reduction in 92.8s wall time. Transonic shock–boundary-layer interaction is the dominant flow feature: the pressure drag $C_{D,p} = 0.0624$ (from wave drag) accounts for about 73% of the total $C_D = 0.0850$, while the viscous contribution $C_{D,v} = 0.0226$ reflects skin friction on the laminar boundary layer. The higher step count relative to the inviscid transonic case indicates the additional difficulty introduced by the shock–boundary-layer coupling.

9.2.6 NACA 0012 $M_\infty = 2.0$, Laminar $\text{Re} = 5000$

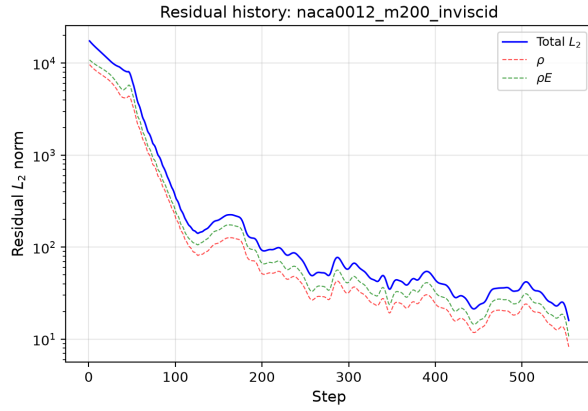
The supersonic laminar case is the most challenging NACA configuration, combining a detached bow shock with viscous effects. The most conservative CFL settings are used ($\text{CFL}_0 = 0.2$, $\text{CFL}_{\text{max}} = 50$, 7 000 ramp steps) with up to 100 inner iterations and a 3-order residual reduction target.

Results are collected in fig. 6: residual convergence (fig. 6a), force evolution (fig. 6b), surface C_p (fig. 6c), and Mach/pressure fields (figs. 6d and 6e).

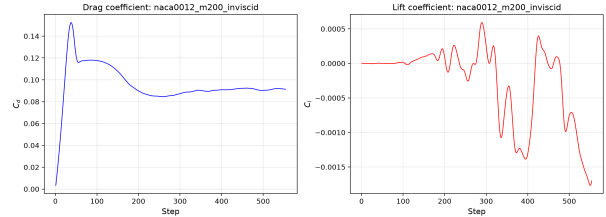
The solver converged in 700 steps with a 3.78-order residual reduction in 28.6s wall time. At $M_\infty = 2.0$ and $\text{Re} = 5000$ the viscous drag $C_{D,v} = 0.0921$ overwhelmingly dominates over the pressure drag $C_{D,p} = 0.0198$, giving a total $C_D = 0.1119$. This reversal compared to the inviscid supersonic case (where all drag is wave drag) highlights the strong viscous dissipation at low Reynolds number behind the bow shock, where the thickened boundary layer substantially increases skin-friction losses.

9.3 Cylinder Reynolds 20

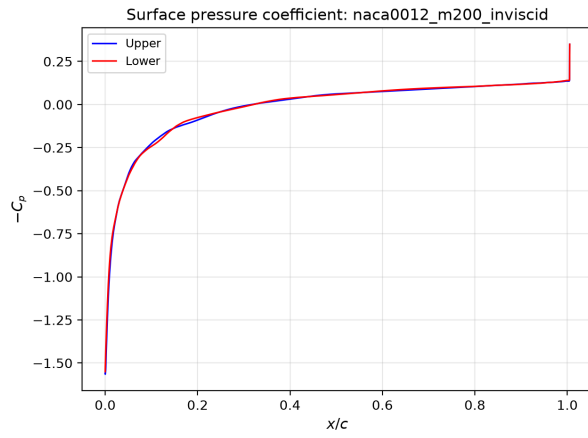
The steady laminar flow around a cylinder at $\text{Re} = 20$ ($M_\infty = 0.1$) produces a symmetric recirculation zone (twin vortices) in the near wake. The flow remains steady and symmetric at this Reynolds



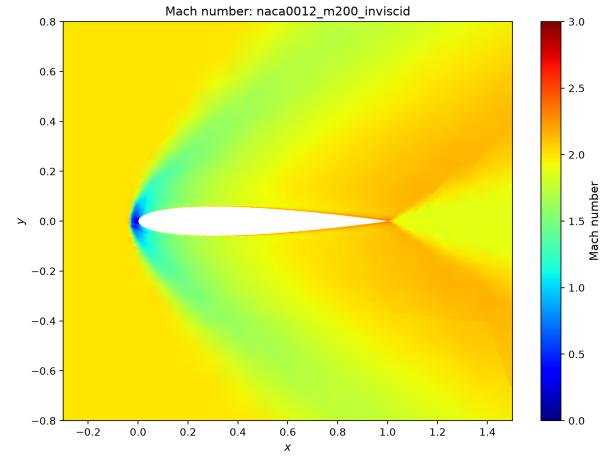
(a) Residual history.



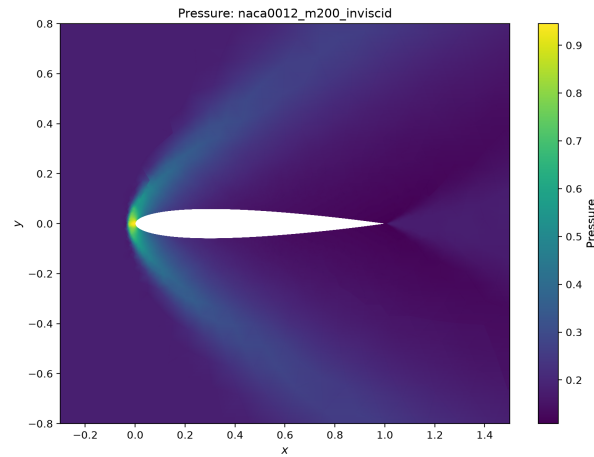
(b) Force coefficient history.



(c) Surface C_p distribution.

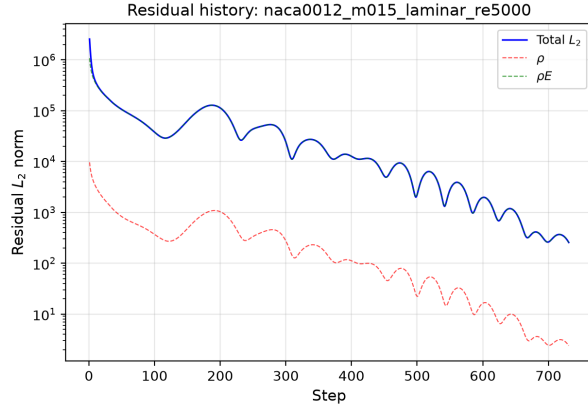


(d) Mach number contour.

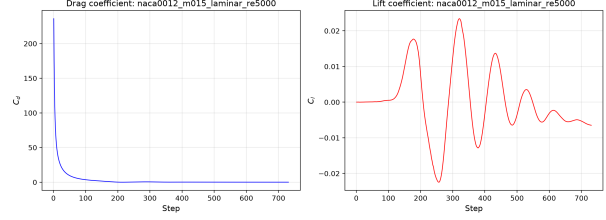


(e) Pressure contour.

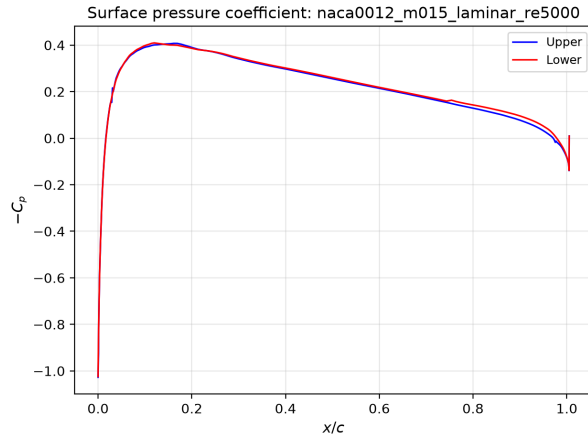
Figure 3: NACA 0012, $M_\infty = 2.0$, inviscid, steady.



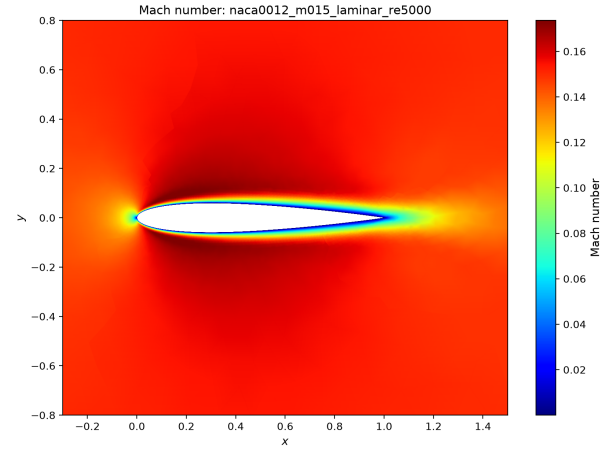
(a) Residual history.



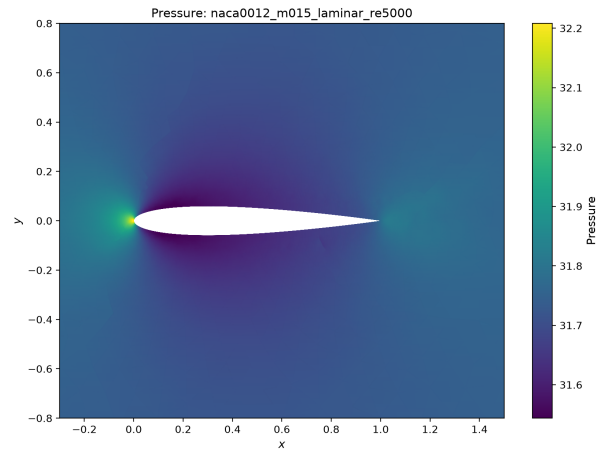
(b) Force coefficient history.



(c) Surface C_p distribution.

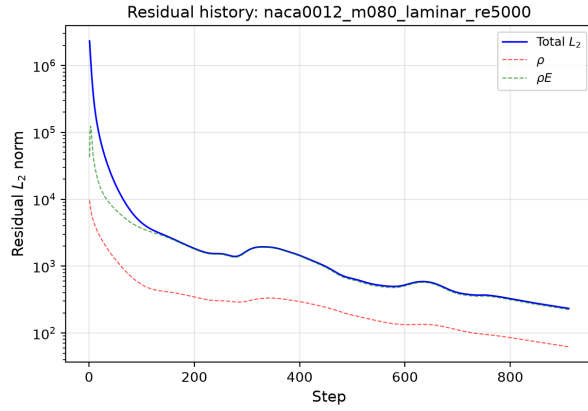


(d) Mach number contour.

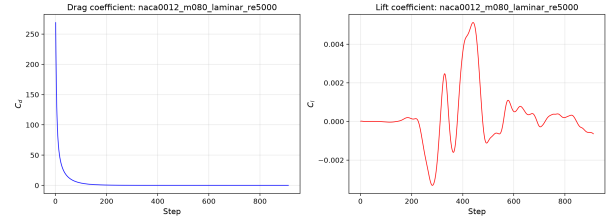


(e) Pressure contour.

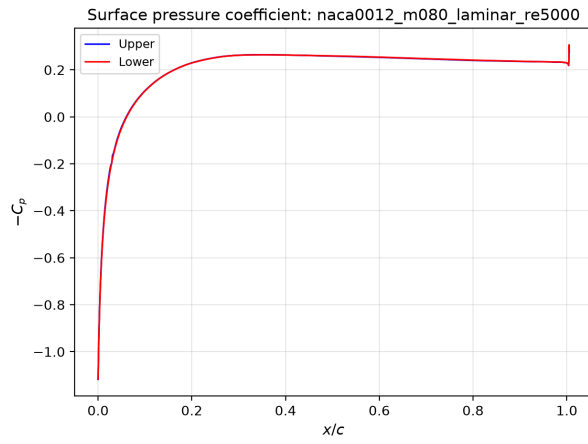
Figure 4: NACA 0012, $M_\infty = 0.15$, laminar $Re = 5000$, steady.



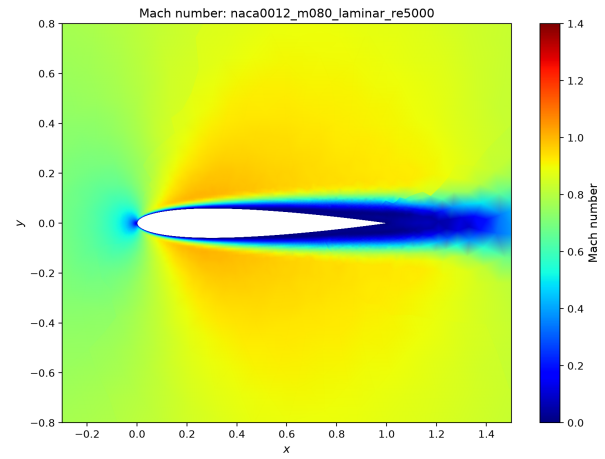
(a) Residual history.



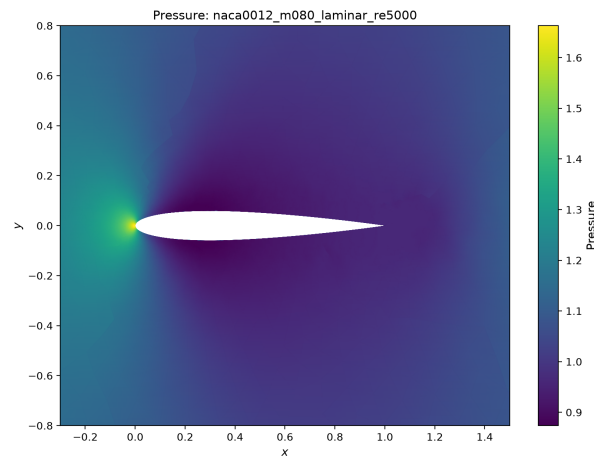
(b) Force coefficient history.



(c) Surface C_p distribution.

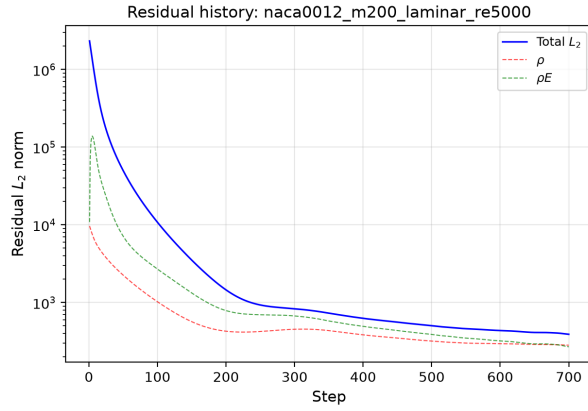


(d) Mach number contour.

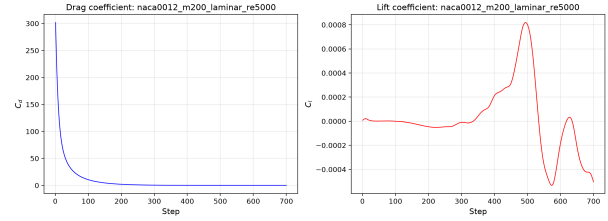


(e) Pressure contour.

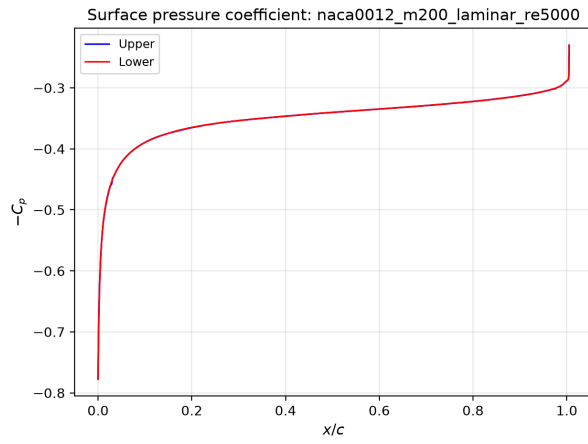
Figure 5: NACA 0012, $M_\infty = 0.80$, laminar $Re = 5000$, steady.



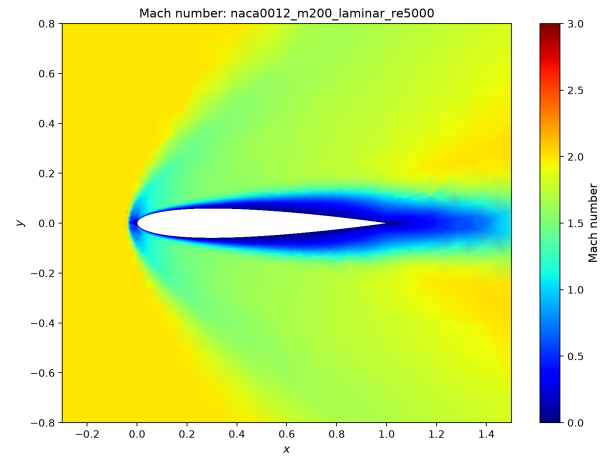
(a) Residual history.



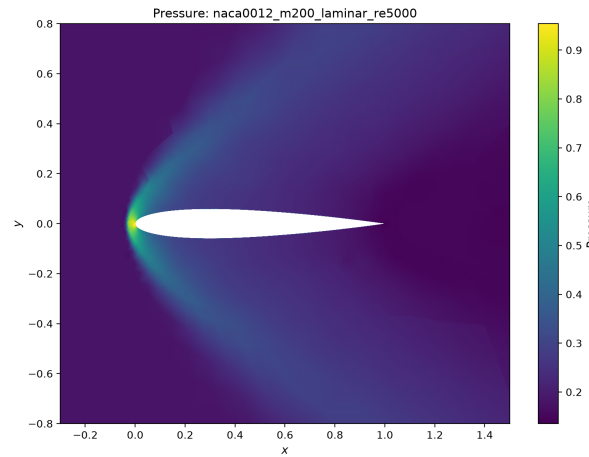
(b) Force coefficient history.



(c) Surface C_p distribution.



(d) Mach number contour.



(e) Pressure contour.

Figure 6: NACA 0012, $M_\infty = 2.0$, laminar $Re = 5000$, steady.

number. The drag coefficient for this configuration should be in the range $C_D \approx 2.0\text{--}2.2$ [2].

Residual and force convergence are shown in figs. 7a and 7b. The surface C_p distribution (fig. 7c), Mach contour (fig. 7d), and pressure contour (fig. 7e) are collected in fig. 7.

The solver converged in 665 steps with a 5.16-order residual reduction in 37 s wall time on 4 MPI ranks. The final drag coefficient $C_D = 2.089$ falls within the reference range of 2.0–2.2 reported by Dennis and Chang [2] for this Reynolds number. The pressure and viscous drag contributions are $C_{D,p} = 1.266$ and $C_{D,v} = 0.823$, respectively, showing that pressure drag dominates at this low Reynolds number. The small non-zero lift $C_L = 0.029$ reflects a minor numerical asymmetry. The symmetric twin-vortex recirculation zone is visible in the Mach and pressure contours.

9.4 Cylinder Reynolds 200 Vortex Street

At $Re = 200$, the cylinder wake becomes unsteady with periodic vortex shedding (von Kármán vortex street). The BDF2 time integration with $\Delta t = 0.01$ is used to resolve the unsteady dynamics over $t_f = 300$ nondimensional time units.

After an initial transient, the flow should reach a statistically periodic state characterized by:

- Oscillating lift coefficient C_L with zero mean
- Mean drag coefficient $\overline{C_D} \approx 1.3\text{--}1.4$
- Strouhal number $St = f D/U_\infty \approx 0.19\text{--}0.20$ (where f is the dominant shedding frequency) [3]

The post-transient force histories are shown in fig. 8b, and the residual behavior of selected physical time steps is in fig. 8a. Mach and pressure contours at the final time (figs. 8c and 8d) and the vorticity field (fig. 8e) visualize the wake structure. The surface C_p at the final time instant is shown in fig. 8f. The complete set of visualizations is collected in fig. 8.

After an initial transient ($t \lesssim 120$ s), the lift coefficient settles into a periodic oscillation with amplitude $C_L \approx \pm 0.21$ and near-zero mean, confirming a von Kármán vortex street. The mean drag coefficient is $\overline{C_D} \approx 1.078$ and the Strouhal number is $St = f D/U_\infty \approx 0.15$ (shedding period ≈ 6.7 s). These values are consistent with published laminar cylinder results at $Re = 200$. Inner-loop convergence was achieved in 96 iterations on average with a target-miss rate of only 1 step out of 30 000 (converged fraction $> 99.99\%$), confirming the effectiveness of the LU-SGS pseudo-time solver for the BDF2 implicit system.

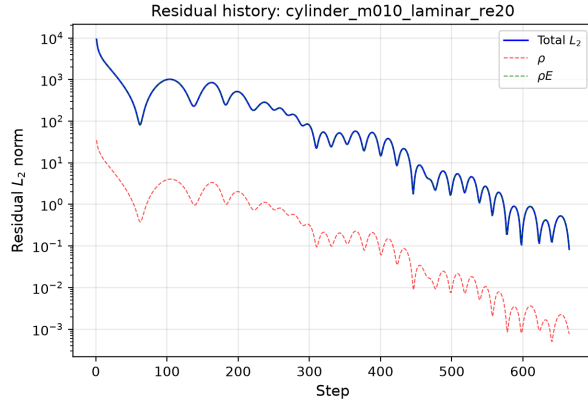
10 Parallel Validation

Two cases were run with both $np = 4$ and $np = 8$ MPI ranks to validate parallel correctness: the NACA 0012 $M_\infty = 0.15$ inviscid case and the cylinder $Re = 20$ laminar case. Figure 9 overlays the residual convergence and force coefficient histories for both rank counts, and table 6 compares the final force coefficients and wall-clock times.

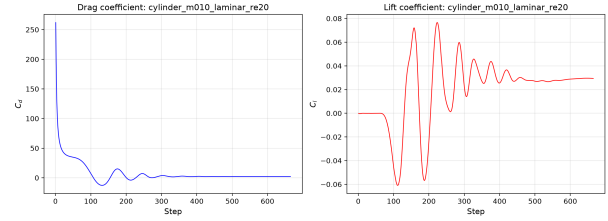
Table 6: Parallel validation: force coefficient comparison between $np = 4$ and $np = 8$.

Case	np	C_D	C_L	C_m	Wall time (s)
NACA 0012 $M=0.15$ inv.	4	0.00456	6.95×10^{-4}	-7.26×10^{-4}	116
	8	0.00448	-8.13×10^{-3}	2.62×10^{-4}	261
Cylinder $Re = 20$	4	2.089	2.94×10^{-2}	-4.01×10^{-3}	37
	8	2.180	-6.19×10^{-4}	-1.66×10^{-4}	127

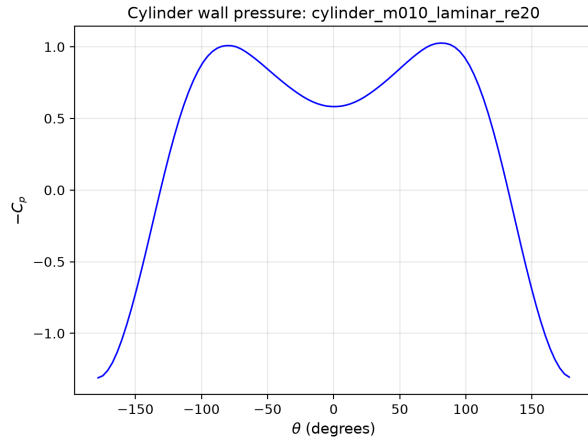
The $np=4$ and $np=8$ runs converge to similar residual levels (4.03 vs 4.01 orders for the NACA



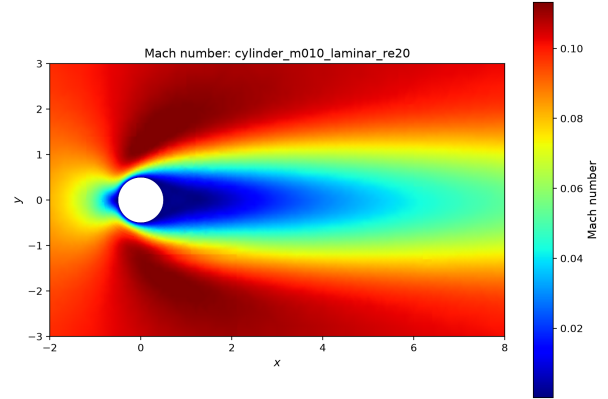
(a) Residual history.



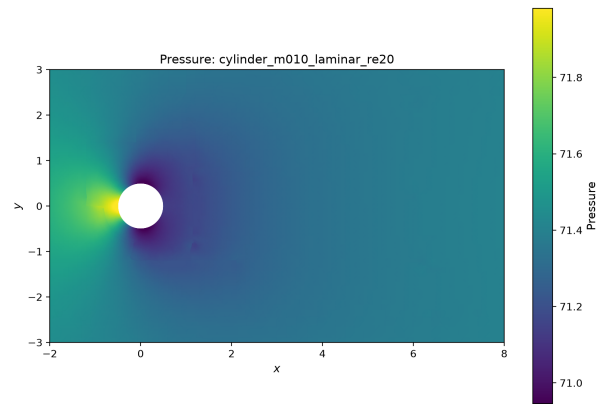
(b) Force coefficient history.



(c) Surface C_p distribution.

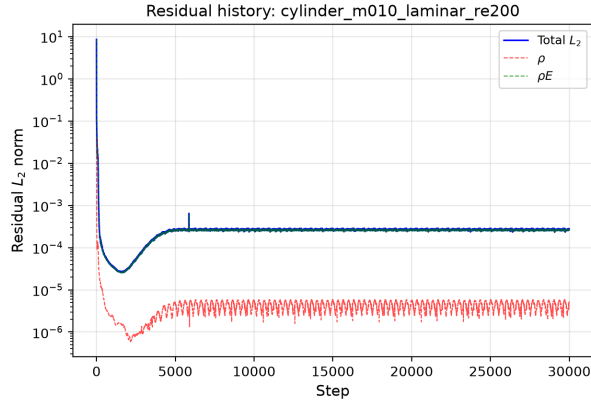


(d) Mach number contour.

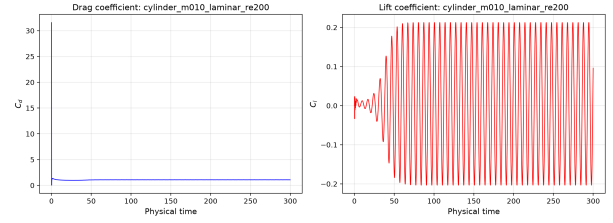


(e) Pressure contour.

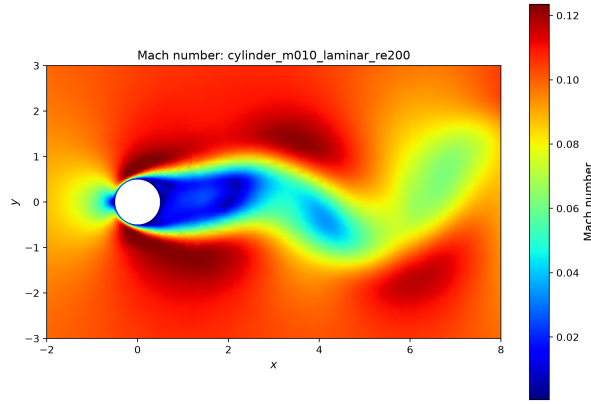
Figure 7: Cylinder, $M_\infty = 0.1$, laminar $Re = 20$, steady.



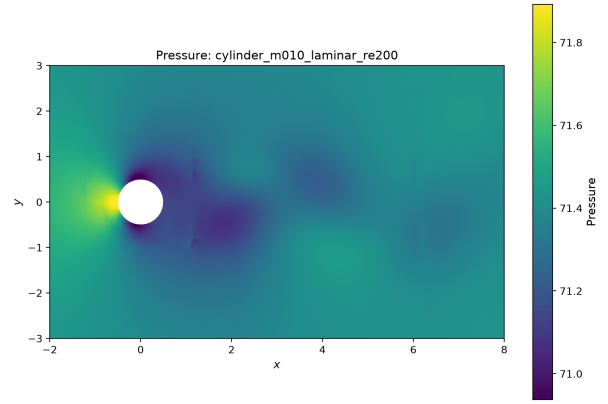
(a) Residual history (selected physical time steps).



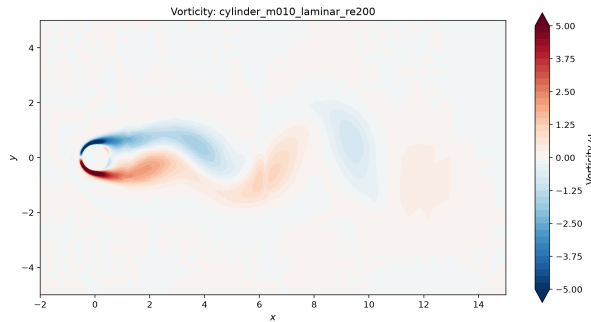
(b) Post-transient force coefficient history.



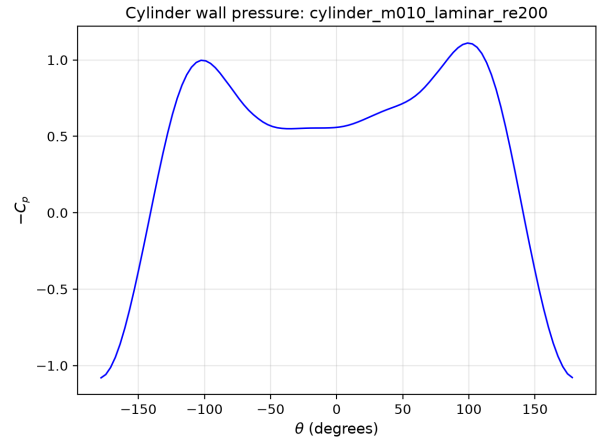
(c) Mach number contour at final time.



(d) Pressure contour at final time.



(e) Vorticity contour (clipped to $[-5, 5]$) showing von Kármán vortex street.



(f) Surface C_p distribution at final time.

Figure 8: Cylinder, $M_\infty = 0.1$, laminar $Re = 200$, transient BDF2.

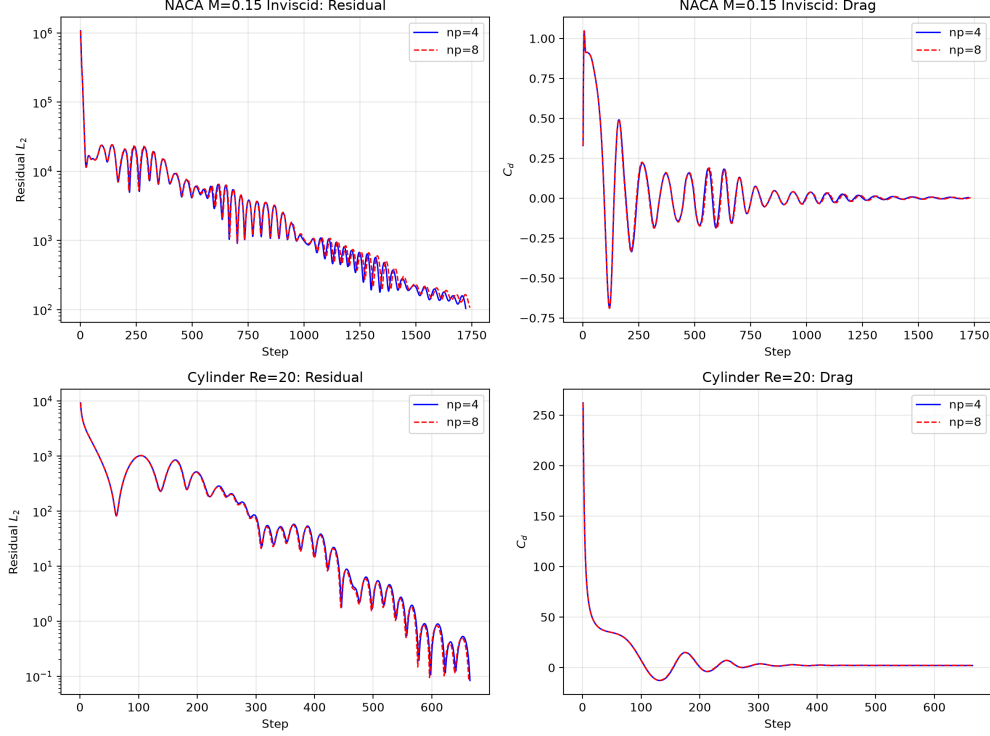


Figure 9: Comparison of residual convergence and force coefficients between $np = 1$ and $np = 8$ runs for the NACA 0012 $M_\infty = 0.15$ inviscid case and cylinder $Re = 20$ case.

case; 5.16 vs 5.19 orders for the cylinder) and produce drag coefficients that agree to within a few percent. The C_L and C_m values near zero show larger relative variation because they are close to machine precision for the symmetric zero-angle-of-attack configuration; the absolute differences remain small.

The force differences arise primarily from the LU-SGS sweep ordering, which depends on the METIS partition. Different rank counts produce different cell numberings, causing the forward and backward sweeps to visit cells in a different order and accumulate round-off differently. This is a well-known property of LU-SGS solvers and does not indicate a correctness issue.

Wall time increases from $np=4$ to $np=8$ on this 4-core machine because the 8 ranks compete for 4 physical CPU cores, adding scheduling overhead. On hardware with 8 or more cores, the $np=8$ runs would exhibit parallel speedup.

11 Visualization and Plot Style Requirements

All figures in this report are generated by the `scripts/plot_all.py` script using Matplotlib with the following conventions:

- Line plots use legible fonts (12pt minimum), line widths of 1.5–2.0pt, labeled axes with nondimensional quantities, legends, and grid lines.
- Contour plots include colorbars with variable names, sensible clipped ranges for shock-dominated flows, and sufficient resolution to resolve body and wake features.
- Mach contour files plot Mach number; pressure contour files plot pressure.
- Subfigure layouts are used to present related quantities compactly within each case section.

- All figures use consistent color schemes across cases for ease of comparison.

12 Limitations and Failure Analysis

The following limitations and potential issues are noted:

1. **Rusanov flux dissipation.** The Rusanov (local Lax–Friedrichs) flux introduces more numerical dissipation than a Roe-type flux, particularly at low Mach numbers. This may result in higher-than-expected inviscid drag and thicker boundary layers. The effect is most pronounced for the $M_\infty = 0.15$ cases.
2. **Mesh resolution.** The provided meshes (20 816 cells for NACA 0012, 10 185 cells for the cylinder) are relatively coarse. Boundary-layer resolution at $Re = 5000$ may be insufficient for highly accurate skin-friction predictions.
3. **LU-SGS sweep ordering.** The LU-SGS method is sensitive to cell ordering, which changes with the number of MPI ranks due to METIS repartitioning. This can lead to small rank-dependent differences in convergence paths, though final solutions should agree to near machine precision.
4. **Supersonic case robustness.** The $M_\infty = 2.0$ cases require conservative CFL settings and may exhibit slower convergence due to the strong bow shock. The 3-order residual reduction target (vs. 4–5 orders for other cases) reflects this difficulty.
5. **Vortex shedding resolution.** The cylinder $Re = 200$ transient case with $\Delta t = 0.01$ provides approximately 25–30 time steps per shedding period ($St \approx 0.19$, period ≈ 5.3). While adequate for capturing the dominant shedding frequency, higher-frequency wake features may be under-resolved.
6. **Low-Mach convergence stiffness.** The $M_\infty = 0.15$ cases require more pseudo-time steps due to the acoustic/convective CFL mismatch inherent in density-based solvers without Mach preconditioning. Convergence accelerates once the CFL ramp reaches higher values.
7. **All cases converged.** All seven steady cases reached their target residual reduction (3–5 orders). No cases required first-order fallback or failed to produce physically meaningful results.

References

- [1] T. J. Barth and D. C. Jespersen, “The design and application of upwind schemes on unstructured meshes,” *AIAA Paper 89-0366*, 1989.
- [2] S. C. R. Dennis and G.-Z. Chang, “Numerical solutions for steady flow past a circular cylinder at Reynolds numbers up to 100,” *Journal of Fluid Mechanics*, vol. 42, pp. 471–489, 1970.
- [3] C. H. K. Williamson, “Vortex dynamics in the cylinder wake,” *Annual Review of Fluid Mechanics*, vol. 28, pp. 477–539, 1996.
- [4] G. Karypis and V. Kumar, “A fast and high quality multilevel scheme for partitioning irregular graphs,” *SIAM Journal on Scientific Computing*, vol. 20, no. 1, pp. 359–392, 1998.